Forecasting and Macroeconomic Analytics

ECON 522

Elena Pesavento — Emory University

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MIDAS
Mixed Data Sampling for Time Series Analysis

Economic Data at Different Frequencies

The Challenge

Economic and financial variables are measured at different frequencies

Low Frequency:

- GDP: Quarterly
- Trade balance: Quarterly
- CPI: Monthly

High Frequency:

- Stock returns: Daily
- Exchange rates: Daily
- Interest rates: Daily

Motivating Question

Forecasting Problem

How can we forecast **quarterly GDP growth** using **daily stock returns**?

The frequency mismatch problem:

- 1 quarter = approximately 65 trading days
- We have 65 data points per quarter for stocks
- But only 1 data point per quarter for GDP

Question

How do we use all the daily information without losing it?

Traditional Approaches (and Their Problems)

1. **Aggregation:** Average daily returns to quarterly
   - × Loses within-quarter dynamics
   - × Treats all days equally
2. **Interpolation:** Create artificial daily GDP data
   - × Creates spurious patterns
   - × GDP doesn't actually change daily
3. **Skip-sampling:** Use only last day of quarter
   - × Discards 99% of the data!

The MIDAS Solution

Key Idea

Use **all** high-frequency data directly to predict low-frequency outcomes

Instead of: 65 daily returns $\rightarrow$ 1 quarterly average $\rightarrow$ forecast GDP

MIDAS does: 65 daily returns $\rightarrow$ weighted combination $\rightarrow$ forecast GDP

Developed by: Ghysels, Santa-Clara, and Valkanov (2004)

quarterly $\bar{\tau}_{daily}$

The MIDAS Framework

Basic Notation

- y_t: Low-frequency variable (e.g., quarterly GDP growth)
- $x_{t-i/m}$: High-frequency variable (e.g., daily stock returns)
- m: Number of high-frequency periods per low-frequency period

Example

If y is quarterly and x is daily:

- $m \approx 65$ trading days per quarter
- $x_{t-1/65}$ = stock return 1 day ago
- $x_{t-65/65}$ = stock return 65 days ago (1 quarter)

The MIDAS Regression Model

General Form

$$y_t = \beta_0 + \beta_1 B(L^{1/m}; \boldsymbol{\theta}) x_t + \varepsilon_t$$

where:

- $L^{1/m}$ = lag operator at high frequency
- $B(L^{1/m}; \boldsymbol{\theta})$ = **weighting polynomial**
- $\boldsymbol{\theta}$ = hyperparameters determining weight shape

Expanded Form

$$y_t = \beta_0 + \beta_1 \sum_{k=1}^{K} w(k; \boldsymbol{\theta}) x_{t-k/m} + \varepsilon_t$$

where $\sum_{k=1}^{K} w(k; \boldsymbol{\theta}) = 1$

t

quarters

1 1960:1

2 1960:2

3 1960:3

4 1960:4

5 1960:5

$t-1/m$

1960:1, month 1

month 2

m_3

m_4

y_t

$y \; X_{t-1/n}$

$y_t =$

$B_0 + B_1 W_1 X_{t-1/m} +$

$B_1 W_2 X_{t-2/m} +$

$B_1 W_3 X_{t-3/m}$

B_1 captures relationship between financial data and GDP

Why Weighted Lags?

The Problem

If we include 65 daily lags, we need to estimate 65 separate coefficients!

The MIDAS Solution

Specify a *functional form* for how weights decline with lag length

Benefits:

1. **Parsimony:** Estimate only 2–3 hyperparameters instead of K parameters
2. **Smoothness:** Weights vary smoothly $\rightarrow$ reduces overfitting
3. **Flexibility:** Can capture different lag patterns

Weighting Schemes

Exponential Almon Lag Weighting

Most popular scheme (Ghysels et. al 2007)

$$w(k;\theta_1,\theta_2) = \frac{\exp(\theta_1 k + \theta_2 k^2)}{\sum_{j=1}^{K} \exp(\theta_1 j + \theta_2 j^2)}$$

Parameters Control Shape

- $\theta_1 < 0$: Weights decline with lag length
- $\theta_2 < 0$: Hump-shaped weights (rise then fall)
- $\theta_2 = 0$: Exponential decay
- $\theta_1, \theta_2 > 0$: Weights increase with lag

Example Weight Patterns

Different parameter combinations yield different patterns:

Exponential decay:

- $\theta_1 = -0.1$
- $\theta_2 = 0$
- Recent lags matter most

Hump shape:

- $\theta_1 = 0.05$
- $\theta_2 = -0.01$
- Middle lags matter most

Key Point

The data determines which pattern fits best!

Beta Weighting Scheme

Alternative scheme using Beta distribution:

$$w(k;\theta_1,\theta_2) = \frac{(k/K)^{\theta_1-1}(1-k/K)^{\theta_2-1}}{\sum_{j=1}^{K}(j/K)^{\theta_1-1}(1-j/K)^{\theta_2-1}}$$

Advantages

- Weights automatically stay in $[0,1]$
- Very flexible shapes
- Well-understood statistical properties

Example Application

Example: Forecasting GDP Growth

Research Question

Can daily stock returns predict quarterly GDP growth?

MIDAS Model:

$$y_t = \beta_0 + \beta_1 \sum_{k=1}^{65} w(k; \theta_1, \theta_2) r_{t-k/65} + \varepsilon_t$$

where:

- y_t = quarterly GDP growth rate
- $r_{t-k/65}$ = stock return k days ago
- Weights follow exponential Almon scheme

Interpreting the Parameters

β_1 (Slope Coefficient)

Overall effect of stock returns on GDP growth

- If $\beta_1 > 0$: Higher returns $\rightarrow$ higher GDP growth
- Magnitude shows economic significance

$\boldsymbol{\theta} = (\theta_1, \theta_2)$ (Hyperparameters)

Shape of the lag structure

- $\theta_1 < 0$: Recent returns matter more
- $\theta_2 < 0$: Middle-of-quarter returns peak importance

Visual Interpretation

Example estimated weights:

Suppose we estimate: $\hat{\theta}_1 = -0.05$, $\hat{\theta}_2 = -0.001$

- Weight on yesterday's return: 0.025
- Weight on 1-week-ago return: 0.020
- Weight on 1-month-ago return: 0.012
- Weight on 2-months-ago return: 0.005

Interpretation

Recent returns are weighted most heavily in predicting GDP

Estimation

How to Estimate MIDAS Models

Method: Nonlinear Least Squares (NLS)

Minimize:

$$\min_{\beta_0,\beta_1,\boldsymbol{\theta}} \sum_{t=1}^{T} \left(y_t - \beta_0 - \beta_1 \sum_{k=1}^{K} w(k;\boldsymbol{\theta}) x_{t-k/m} \right)^2$$

theta minimizes this

Why nonlinear?

- Parameters $\boldsymbol{\theta}$ enter nonlinearly through $w(k;\boldsymbol{\theta})$
- Need numerical optimization (e.g., Newton-Raphson)

Estimation Steps

1. **Choose weighting scheme**
   - Exponential Almon, Beta, or other
2. **Choose maximum lag K**
   - How many high-frequency periods to include?
   - Trade-off: more lags vs. parameter uncertainty
3. **Set starting values**
   - Important for convergence
4. **Optimize via NLS**
   - Find $\hat{\beta}_0, \hat{\beta}_1, \hat{\boldsymbol{\theta}}$
5. **Compute standard errors**
   - Use robust/HAC standard errors

Advantages and Extensions

Advantages of MIDAS

1. **Information preservation**
   - Uses all available high-frequency data
2. **Flexible dynamics**
   - Captures complex lag structures
3. **Nowcasting capability**
   - Update forecasts as new high-frequency data arrives
4. **No artificial data**
   - No interpolation or arbitrary aggregation
5. **Parsimonious**
   - Few parameters despite many lags

Limitations

Challenges

1. Model specification
   - Must choose weighting scheme
   - Results can be sensitive to choice
2. Computational intensity
   - Nonlinear optimization can be slow
   - Convergence issues possible
3. Stability assumption
   - Assumes weight pattern stable over time
   - May not hold during structural breaks

Extensions of MIDAS

Recent developments (Andreou et al. 2013, Ghysels and Andreou 2018):

- **U-MIDAS:** Unrestricted MIDAS
  - Uses regularization (ridge, LASSO)
  - More flexible but less parsimonious (Foroni et al 2015)
- **ADL-MIDAS:** Autoregressive Distributed Lag
  - Includes lags of dependent variable
- **Multi-frequency MIDAS**
  - Combine daily, weekly, monthly predictors
- **Nonlinear MIDAS**
  - Threshold effects, time-varying parameters

Summary

Key Takeaways

What is MIDAS?

A regression framework for mixing data at different sampling frequencies

1. Uses **weighted distributed lags** with parsimonious parameterization
2. Common weighting schemes: **Exponential Almon** and **Beta**
3. Estimated via **nonlinear least squares**
4. Particularly useful for **forecasting and nowcasting**
5. Preserves high-frequency information without aggregation

Applications in Practice

MIDAS is widely used for:

- GDP nowcasting using financial indicators
- Inflation forecasting with commodity prices
- Volatility prediction using high-frequency returns
- Central bank policy analysis
- Macroeconomic surveillance

Real-World Example

Federal Reserve uses MIDAS-type models for GDP nowcasting (GDPNow)